

Wakeup/Shutdown

Modes Of Operation

The MAX17335 supports six power modes (three active modes and three shutdown modes) as shown in [Table 9](#) with descriptions of the features available in each mode, the typical current consumption of each mode, and the method to enter and exit each mode.

Table 9. Modes of Operation

MODE	SYMBOL	CONSUMPTION (TYPICAL) (μA)	DESCRIPTION
Active	IQ _A	35	Full Functionality. The Protection FETs, charge pump, and ADC are on. Tasks execute every 351ms.
Hibernate (optional)	IQ _H	21	FETs, charge pump, and ADC is on. Tasks execute every 1.4s. If enabled, the the device automatically enters and exits this mode depending on current measurements. Entering hibernate mode requires a low enough current for a long enough duration. Exiting requires just one high enough current event. For specific details regarding the thresholds, see the nHibCfg register definition.
Protected and Awake	IQ _P	11	ADC is on. The FETs and charge pump are disabled due to a protection fault, disconnecting the battery from the system. RAM is preserved and the gauge continues to monitor the battery. Firmware remains awake and ready to communicate or enable the battery. Firmware executes every 1.4s.
Ship*	IQ _S	11	Similar state as "Protected and Awake" except the firmware is responsive to wakeup events such as: charger connection, communications wakeup, or pushbutton wakeup (depending on which wakeups are enabled by configuration). Firmware executes every 1.4s.
		8	Similar state as "Protected and Awake" except the firmware is responsive to wakeup events such as: charger connection, communications wakeup, or pushbutton wakeup (depending on which wakeups are enabled by configuration). Firmware executes every 5.625s.
DeepShip*	IQ _{DS}	0.5	FETs, charge pumps, ADC, and firmware are all placed into a shutdown state. The only activity alive relates to analog circuits that monitor for wakeup conditions (charger detection, communications, or pushbutton, depending on which are enabled).
DeepShip2*/ Undervoltage Shutdown	IQ _{UV}	0.5	FETs, charge pumps, ADC, firmware, and most wakeup circuits are powered down. Only the charger detection wakeup circuit remains powered in this mode to best conserve the small remaining battery capacity and prevent deep discharge.

*When an I²C SHIP command (setting Config.SHIP = 1) or I²C SCL/SDA lines collapse (and depending on whether COMMSH is enabled), the MAX17335 either enters Ship (if nProtCfg.DeepShpEn = 0) or DeepShip (if nProtCfg.DeepShpEn = 1) or DeepShip2 (if nProtCfg.DeepShp2En = 1) according to the configuration.

Table 10. MAX17335 Ship Modes

	ENTER	WAKEUP	FUNCTIONALITY	nProtCfg. DeepShipEn	nProtCfg. DeepShip2En
8µA Ship	Config.SHIP or SDA- collapse	I²C, Pushbutton, or Charge Source	5.6s Measurements/ Updates	0	0
0.5µA DeepShip			No updates	1	0
0.5µA DeepShip2		Charge Source		0	1
0.5µA UVShdn	VCell < UVShdn			X	X

The MAX17335 can be awoken with a variety of methods depending on the configuration. If pushbutton wake-up is enabled (nConfig.PBen = 1), then consistently pulling the ALRT/PIO pin low, either by pushbutton or system configuration, wakes up the device. A high-to-low transition on any of the communication lines wakes up the device. A consistent connection to a charge source wakes up the device.

The MAX17335 prevents accidental wakeup when the system is boxed and shipped. When awoken by the pushbutton from ShipMode, the IC debounces the pushbutton signal to ensure that the wakeup is valid. If no valid wakeup is discovered, the device returns to Ship Mode.

Note: There is no debounce when exiting from DeepShip or DeepShip2.

The I_Q in the active, hibernate, and ship modes are impacted by the configuration of the IC. [[Recommended nHibCfg Settings and the Impact on I_Q]] shows the recommended configuration settings for the nConfig register and the impact those settings have on the I_Q of each mode. Note that when in hibernate mode, the protection for overtemperature and overvoltage is delayed by the nHibCfg.HibScalar value. It is not recommended to have hibernation enabled with the nHibCfg.HibScalar set to more than 1.4 seconds.

Table 11. Recommended nHibCfg Settings and the Impact on I_Q

AVAILABLE LOW POWER CONFIGURATION	nHibCfg	FETS- OFF	FETS-ON MODES	UPDATE RATE		NOTES
		SHIP I _Q (μA)	ACTIVE/ HIBERNATE I _Q (μA)	ACTIVE (s)	SHIP (s)	
1.4s Ship	0x0909	10	24/NA	0.351	1.4	—
1.4s Ship + Hibernate	0x8909	10	24/18	0.351	1.4	Overtemperature and overvoltage detection is delayed by 1.4s when in hibernate mode.
5.625s Ship	0x090B	7	24/NA	0.351	5.625	—